DIY Night Light: STEM Lesson Plan

Planrs Academy | Session 1: Grade 1

1. Lesson Overview

This lesson introduces young learners to the basics of electronics and light physics through a hands-on craft project. Students will learn how to build a functional circuit and explore how light interacts with different materials [cite: 1, 2].

> **The Story of Priya:** Small Priya was scared of the dark at night, but she found comfort in the glow of Deepavali diyas and Jugnu fireflies. Today, we help Priya "let that light come" by making our own night lights [cite: 4, 5]!

2. Learning Objectives

- Understand the basic components of a simple circuit (Battery and LED) [cite: 8].
- Identify the flow of electricity in a circular path [cite: 10].
- Classify materials as transparent, translucent, or opaque [cite: 12].
- Practice creative reuse and upcycling of household materials [cite: 24, 30].

3. Science Concepts

How Electricity Flows

Electricity travels from the battery to the LED to create light [cite: 10]. This journey happens in a circle called a **Circuit** [cite: 10].

- **The Power Source:** The battery gives power to the LED [cite: 8].
- **The Connection:** The LED has a long leg (Positive +) and a short leg (Negative -) [cite: 8, 31, 32]. For the light to glow, the legs must touch the correct sides of the battery [cite: 8, 34].

Light Behavior

Light behaves differently depending on what it hits [cite: 12, 16]:

Material Type	Behavior	Example
Transparent	Lets light pass through clearly.	Sweet Box [cite: 13]
Translucent	Lets some light through (soft glow).	Paper Chai Cup [cite: 14]
Opaque	Blocks light completely.	Jar or Opaque Jar [cite: 15]

4. Safety First!

Always follow these tips to keep our workshop safe, *bachche* [cite: 20]:

- **Insulation:** Always cover wire connections with tape for safety [cite: 21, 50].
- **No Heavy Tools:** Do not use big batteries or dangerous tools [cite: 22].
- **Cleanliness:** Keep your hands dry and clean while working with electronics [cite: 23].

5. Materials Checklist

We will use upcycled materials to create our magic [cite: 24, 30]!

- Old Horlicks Jar [cite: 25]
- Transparent Sweet Box [cite: 26]
- Paper Teacup [cite: 28]
- Rangoli Paper [cite: 27]
- Colorful Stickers [cite: 29]
- LED & Battery [cite: 41]
- Electrical Tape [cite: 21]

6. Step-by-Step Construction

Step 1: **Test Your LED**

Press the long and short legs of the LED to the battery. If it glows, it’s ready to shine [cite: 33, 34]!

Step 2: **Decorate Your Jar**

Stick colorful patterns like Diwali Rangoli patterns on your jar. Use stickers and paper cut-outs to make it yours [cite: 37, 38, 39].

Step 3: **Fix the Light Inside**

Place the LED and battery inside the jar. Tape the connections tightly and close the lid neatly [cite: 40, 41]. Attach it carefully [cite: 42]!

Step 4: **The Magic Moment**

Turn off the lights and watch your night light glow! How beautiful [cite: 43, 44]!

7. Quiz Time!

Q1: Which material lets light shine through? [cite: 45]
A) Opaque Jar
B) Paper Tea Cup
C) Transparent Sweet Box [cite: 46]

Q2: What gives energy to the LED to make it glow? [cite: 47]
A) The Paper
B) The Battery [cite: 8, 48]
C) The Jar

Q3: Which is safe to touch? [cite: 49]
A) Untaped Wires
B) Taped Wires [cite: 50]

8. Conclusion & Show and Tell

Great job! Now, show your night light to Daadi and your family. It will keep you safe and happy through the night [cite: 51]. Remember to clean up your workspace together [cite: 52]!